

HYDROGENATED NBR: PATENT LANDSCAPE FOR NBR PREPARATION, DEGRADATION, HYDROGENATION, AND PROCESS INTENSIFICATION

1. ABSTRACT

This report synthesizes a patent set focused on hydrogenated nitrile butadiene rubber (HNBR) and closely related enabling technologies for its manufacture. The core target is NBR-derived material with reduced residual unsaturation, typically achieved by selective hydrogenation of the butadiene-derived double bonds in NBR copolymers. Across the selected patents, the landscape separates into three technical themes: (1) NBR feedstock preparation and molecular-weight control, (2) hydrogenation reactor and catalyst engineering, and (3) post-hydrogenation handling or downstream property optimization. The feedstock patents emphasize solution or emulsion polymerization of acrylonitrile with butadiene, optionally with additional copolymerizable monomers, and use molecular-weight modifiers such as mercaptans in solution polymerization or ultrasound-based degradation to obtain narrower molecular-weight distributions. The hydrogenation patents focus on continuous or batch hydrogenation of unsaturated polymers in the presence of homogeneous or heterogeneous catalysts, with recurring catalyst families including rhodium, ruthenium, osmium, iridium, platinum, palladium, nickel, copper, and titanium. Several disclosures specifically address HNBR by hydrogenating NBR in organic solvent, often after coagulation and redissolution of the NBR latex, while others improve reactor design through multistage agitated reactors or continuous processing to manage exotherm, mass transfer, and product consistency. A separate set of patents addresses metathesis-based molecular-weight reduction before hydrogenation, using ruthenium/osmium carbene catalysts and salt additives to improve catalyst activity, thereby enabling lower molecular weight HNBR with lower Mooney viscosity and narrower polydispersity. The overall formulation/process trend is toward lower residual unsaturation, lower viscosity, narrower molecular-weight distribution, and more controllable industrial processing. The evidence supports a practical synthesis route in which NBR is prepared with controlled molecular weight, optionally degraded by metathesis or physical treatment, then hydrogenated in solution using selective homogeneous catalysts under continuous or well-mixed conditions, followed by coagulation strategies that preserve product quality and facilitate isolation.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US7919563B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 2

- Patent Number: US8957161B2
- Patent Title: Process for reacting chemical compounds in the presence of catalyst systems
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 3

- Patent Number: EP2611837B1
- Patent Title: Method for producing nitrile rubbers in organic solvents
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

Patent 4

- Patent Number: US7737233B2
- Patent Title: Catalyst systems and their use for metathesis reactions
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

Patent 5

- Patent Number: US7932346B2
- Patent Title: Synthetic rubber with narrow molecular weight distribution, a process for its ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 6

- Patent Number: EP3708590B1
- Patent Title: Method for manufacturing hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2022
- Legal Status: Unknown
- Polymer Type: Not disclosed

- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

Patent 7

- Patent Number: US11142708B2
- Patent Title: Processes and systems for petrochemical production integrating deep ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

Patent 8

- Patent Number: EP2147933B1

- Patent Title: Process for the continuous hydrogenation of carbon-carbon double bonds in an ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2012
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

Patent 9

- Patent Number: US10815320B2
- Patent Title: Functionalized resin having a polar linker
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

Patent 10

- Patent Number: US11262338B2
- Patent Title: Modified resins and uses thereof
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2022
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US7919563B2 and US8957161B2 both use metathesis-based degradation of nitrile rubber before or during hydrogenation to reach lower molecular weight HNBR, but US7919563B2 emphasizes low molecular weight and narrow molecular-weight distribution via metathesis in the presence of Ru/Os compounds and optional co-olefin/solvent, whereas US8957161B2 focuses on catalyst-system enhancement by adding salts to metathesis catalysts to increase activity.
- US7737233B2 is a catalyst-system patent closely aligned with US8957161B2, but it specifies alkaline earth metal chlorides, especially calcium chloride or magnesium chloride, as additives

to metathesis catalysts; the shared theme is catalyst activation for nitrile-rubber metathesis, with the later patent family explicitly addressing the problem of catalyst efficiency and reduced catalyst loading.

- EP2611837B1 differs from the metathesis patents by targeting feedstock preparation rather than post-polymer degradation: it discloses free-radical solution polymerization of butadiene/acrylonitrile with molecular-weight modifiers such as mercaptans, mercapto alcohols, disulfides, allyl compounds, aldehydes, halohydrocarbons, thiourea, and saccharin in polar organic solvent, followed by hydrogenation if desired.
- US7932346B2 is not a hydrogenation catalyst patent but a physical degradation route: ultrasound is used to reduce the molecular weight of synthetic rubbers including NBR and HNBR, producing lower Mw and narrower polydispersity without changing the polymer class by chemical degradation.
- EP2147933B1 and EP3708590B1 both address hydrogenation process engineering for NBR-to-HNBR conversion, but EP2147933B1 focuses on continuous hydrogenation in a multistage agitated reactor with homogeneous or heterogeneous catalysts, while EP3708590B1 focuses on a two-step coagulation strategy around solution hydrogenation, using a halogen atom-free metal salt before hydrogenation and a divalent metal salt after hydrogenation to improve isolation and product quality.
- US11142708B2 is not an NBR patent; it is included in the manifest but concerns deep hydrogenation of middle distillates for petrochemical production, so it is technically unrelated to HNBR synthesis despite the shared hydrogenation terminology.
- US10815320B2 and US11262338B2 are resin-modification patents rather than NBR synthesis patents; they are relevant only as downstream rubber-compounding context and do not disclose NBR monomer ratios, hydrogenation conditions, or HNBR preparation routes.
- Across the primary HNBR-relevant patents, the recurring industrial pattern is: prepare NBR with controlled molecular weight, reduce or tailor molecular weight if needed, hydrogenate the diene unsaturation in solution using selective catalysts, and isolate the product by coagulation or continuous processing to manage viscosity and residual unsaturation.

4. CONCLUSION

The patent evidence indicates that successful HNBR manufacture depends on controlling both the NBR precursor and the hydrogenation/isolation sequence. The most consistent technical direction is to start from butadiene/acrylonitrile NBR, preferably with molecular-weight control during polymerization or a subsequent controlled degradation step, then hydrogenate the diene-derived double bonds in solution using selective homogeneous catalysts or robust continuous reactor systems. For lower-viscosity HNBR, the strongest evidence supports precursor molecular-weight reduction before hydrogenation, either by metathesis using Ru/Os carbene catalysts with salt activation or by non-chemical reduction such as ultrasound. For hydrogenation itself, continuous multistage agitated reactors and carefully chosen coagulation steps appear advantageous for heat and mass transfer, product consistency, and isolation. The most defensible synthesis recommendation from the disclosed patents is: prepare NBR with acrylonitrile/butadiene as the

base composition, keep molecular weight in a processable range before hydrogenation, hydrogenate in organic solvent with a selective catalyst system, and use coagulation/isolation conditions that minimize contamination and preserve the desired low residual unsaturation and narrow molecular-weight distribution.

5. REFERENCES

- US7919563B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7919563B2/en>
- US8957161B2 | Process for reacting chemical compounds in the presence of catalyst systems | Not disclosed | US | 2015 | <https://patents.google.com/patent/US8957161B2/en>
- EP2611837B1 | Method for producing nitrile rubbers in organic solvents | Not disclosed | EP | 2017 | <https://patents.google.com/patent/EP2611837B1/en>
- US7737233B2 | Catalyst systems and their use for metathesis reactions | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7737233B2/en>
- US7932346B2 | Synthetic rubber with narrow molecular weight distribution, a process for its ... | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7932346B2/en>
- EP3708590B1 | Method for manufacturing hydrogenated nitrile rubber | Not disclosed | EP | 2022 | <https://patents.google.com/patent/EP3708590B1/en>
- US11142708B2 | Processes and systems for petrochemical production integrating deep ... | Not disclosed | US | 2021 | <https://patents.google.com/patent/US11142708B2/en>
- EP2147933B1 | Process for the continuous hydrogenation of carbon-carbon double bonds in an ... | Not disclosed | EP | 2012 | <https://patents.google.com/patent/EP2147933B1/en>
- US10815320B2 | Functionalized resin having a polar linker | Not disclosed | US | 2020 | <https://patents.google.com/patent/US10815320B2/en>
- US11262338B2 | Modified resins and uses thereof | Not disclosed | US | 2022 | <https://patents.google.com/patent/US11262338B2/en>